QAI LABORATORIES
CERTIFICATION TESTING INSPECTION

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CLIENT: **Flex Trim West**
533 South 500 West
Provo, UT 84601

Test Report Number :	TUL0973-4-R1	**Original Issue Date: June 24, 2024**

Revision Date: 28-Jun-24

SAMPLE ID: The client identified the following test material as:
Rigid Cast FR RFRV8

SAMPLING DETAIL: Test samples were submitted to the laboratory directly by the client. No sampling or sample preparation were observed by QAI staff.

DATE OF RECEIPT: Samples were received at QAI facilities on: 06/07/2024

TESTING PERIOD: 06/17/2024

AUTHORIZATION: Testing was authorized by Matt Anderson for proposal 24FB05301 signed 05/30/2024.

TEST REQUESTED: Perform standard flame spread and smoke density developed classification tests on the sample supplied by the Client in accordance with ASTM E84 - 21 "Standard Method of Test for Surface Burning Characteristics of Building Materials". The foregoing test procedure is comparable to UL 723, ANSI/NFPA No. 255, and UBC No. 8-1.

TEST RESULTS:

Flame Spread	**Smoke Developed**
130	**400**

CONCLUSION: When tested in accordance to ASTM E84-21 the material resulted in a Class 'C'. Detailed test results are presented in the subsequent pages of this report

Prepared By

Scott Berry
Lab Supervisor - Fire

Signed for and on behalf of
QAI Laboratories, Inc.

Michael Lowry
Digitally signed by Michael Lowry
Date: 2024.06.28 12:40:46 -05'00'

Michael Lowry
Operations Manager

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SCOPE: This fire-test-response standard is used for the comparative surface burning behavior of building materials is applicable to exposed surfaces such as walls, ceilings and others. The test is conducted with the specimen in the ceiling position with the surface to be evaluated exposed face down to the ignition source. The material, product, or assembly shall be capable of being mounted in the test position during the test. Thus, the specimen shall either be self-supporting by its own structural quality, held in place by added supports along the test surface, or secured from the back side. The purpose of this test method is to determine the relative burning behavior of the material by observing the flame spread along the specimen. Flame spread and smoke developed index are reported. However, there is not necessarily a relationship between these two measurements.

USE: The use of supporting materials on the underside of the test specimen has the ability to lower the flame spread index from those which might be obtained if the specimen could be tested without such support. These test results do not necessarily relate to indices obtained by testing materials without such support.

Testing of materials that melt, drip, or delaminate to such a degree that the continuity of the flame front is destroyed, results in low flame spread indices that do not relate directly to indices obtained by testing materials that remain in place.

This standard is used to measure and describe the response of materials, products, or assemblies to heat and flame under controlled conditions, but does not by itself incorporate all factors required for fire-hazard or fire-risk assessment of the materials, products, or assemblies under actual fire conditions.

PROCEDURE: A brief overview of the method is as follows: The test specimen, a material between 20 and 24 inches in width by 24 feet +/- 12 inches in length is loaded onto the water cooled ledge of the fire test chamber when tested to ASTM E84 or CAN/ULC-S102. If tested to CAN/ULC-S102.2 the specimen is tested on the chamber floor. The inside dimensions are 17 3/4 inches +/- 1/4" wide by 12 inches +/- 1/2" deep by 25 feet long. The fire test chamber is a rectangular horizontal duct with a removable lid. The sides and base of the chamber are lined with an insulated firebrick with pressure tight observation windows down one side for a technician to observe flame progression during the duration of the 10-minute test period. The chamber lid is lowered into test position with non combustible concrete board placed between the specimen and chamber lid. A draft of 240 feet per minute which is maintained inside the test chamber throughout the test period by the means of an electronic fan afterburner and an electronically controlled damper door system located downstream of the test chamber in the exhaust ducting. The test is started when the test flame is ignited at the front of the test chamber. An electronic photocell system located in the exhaust system downstream from the test chamber is used to plot the smoke developed for use in calculating the smoke developed index while a technician plots the flame spread distance used in determining the flame spread index. The test is run for the 10 minute duration in accordance to the method.

(See Diagrams in the Appendix of this report.)

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PREPARATION AND CONDITIONING:

The sample board material was delivered to QAI in 24 inches wide X 5 feet long X 0.25 inches thick pieces. Five of these pieces were used for the test. (See Photos in Appendix of this report). The specimen was placed in the conditioning room (maintained at 70 ± 5° F and a relative humidity of 50 ± 5%) for a minimum of 72 hours prior to testing.

MOUNTING METHOD:

The test ready sample consisting of five pieces measuring 24 inches wide X 5 feet long and an overall test thickness of 0.25 inches were stacked end to end on the chamber ledge to fulfill the chamber requirements for testing. Prior to testing the samples were covered with 1/4 inch cement board as required in the test method.

ASTM E84 TEST RESULTS:

CLIENT NAME:	**Flex Trim West**	**TEST DATE:** 06/17/2024	
SAMPLE ID:	Rigid Cast FR RFRV8		
SAMPLE IGNITION:		**00:22**	Minutes / Seconds
MAX FLAME FRONT:		**18.0**	Feet
TIME TO MAXIMUM SPREAD:		**08:28**	Minutes / Seconds
TEST DURATION:		**10:00**	Minutes / Seconds
SUMMARY:	FLAME SPREAD:	**130**	*132 Unrounded*
	SMOKE DEVELOPED:	**400**	*408 Unrounded*

OBSERVATIONS:

Charring was observed at 00:10. Blistering was observed at 00:11. Flaming Dripping was observed at 01:33. Afterburn was observed at 10:01.

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SUMMARY OF ASTM E84 / UL 723 RESULTS:

Because of the possible variations in reproducibility, the results are adjusted to the nearest figure divisible by 5.
Smoke Density values over 200 are rounded to the nearest figure divisible by 50.

In order to obtain the Flame Spread Classification, the above results should be compared to the following table:

NFPA CLASS[1]	IBC CLASS[2]	FLAME SPREAD	SMOKE DEVELOPED
A	A	0 through 25	Less than or equal to 450
B	B	26 through 75	Less than or equal to 450
C	C	76 through 200	Less than or equal to 450

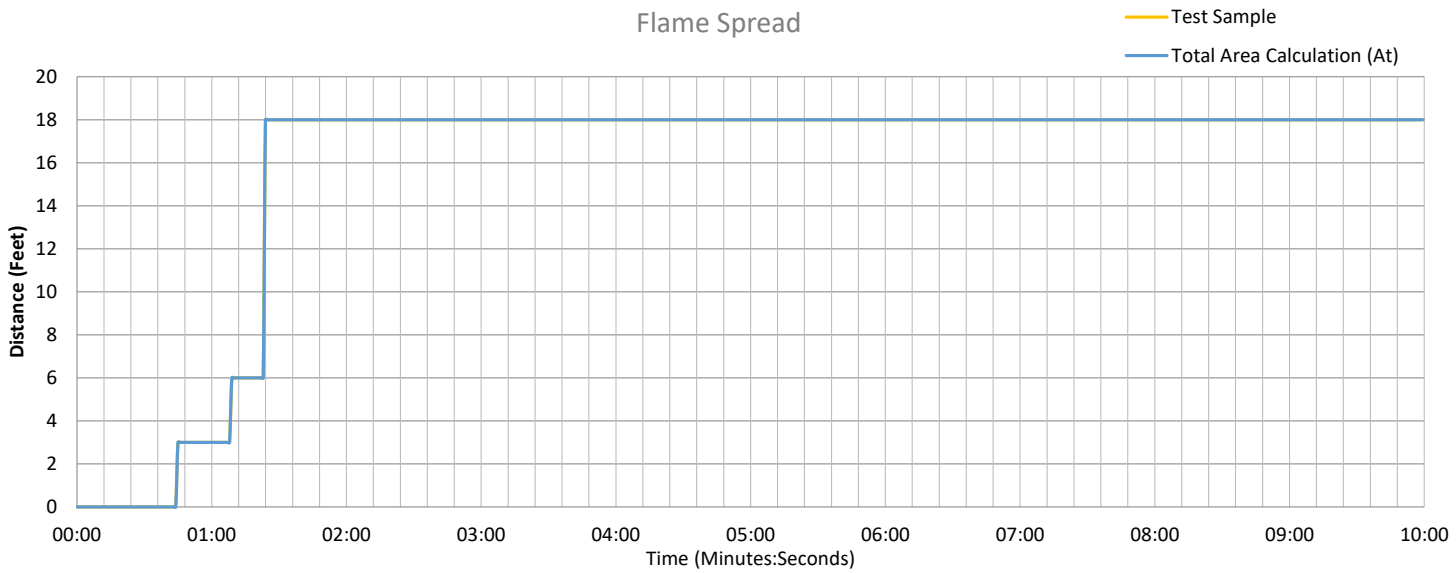
BUILDING CODES CITED:

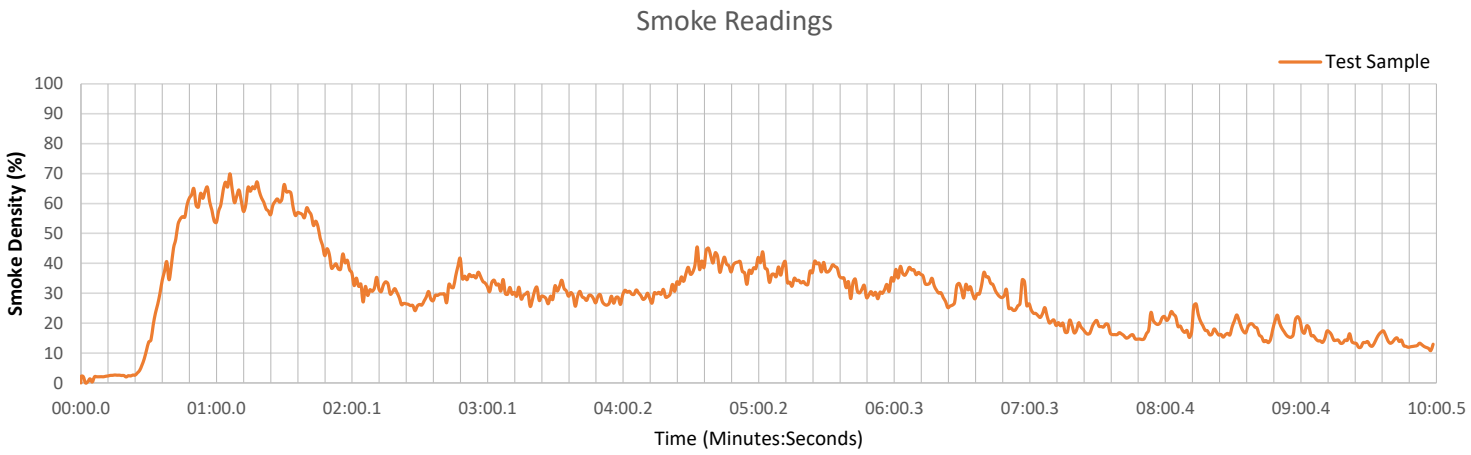
1. National Fire Protection Association, ANSI/NFPA No. 101, "Life Safety Code"

2. International Building Code, Chapter 8, Interior Finishes, Section 803.

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RESULTS CONTINUED:





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APPENDIX

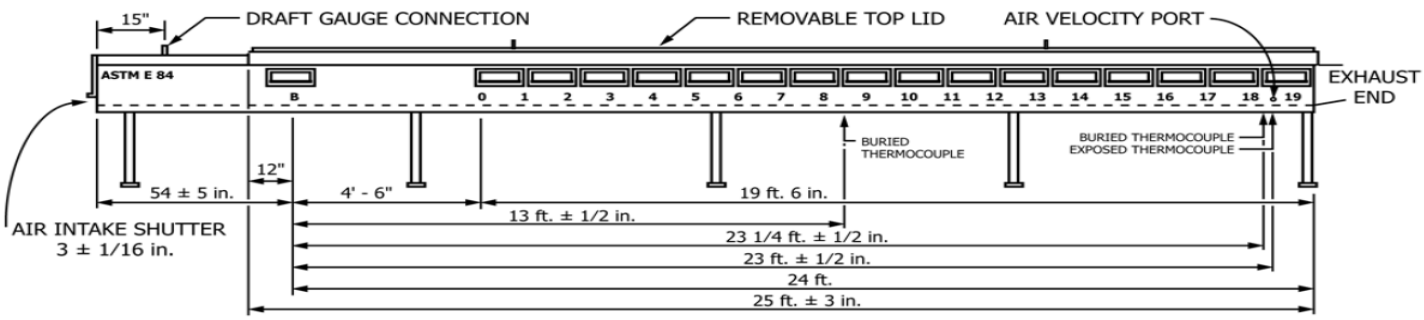


Diagram 1. Test Chamber side view showing critical dimensions.

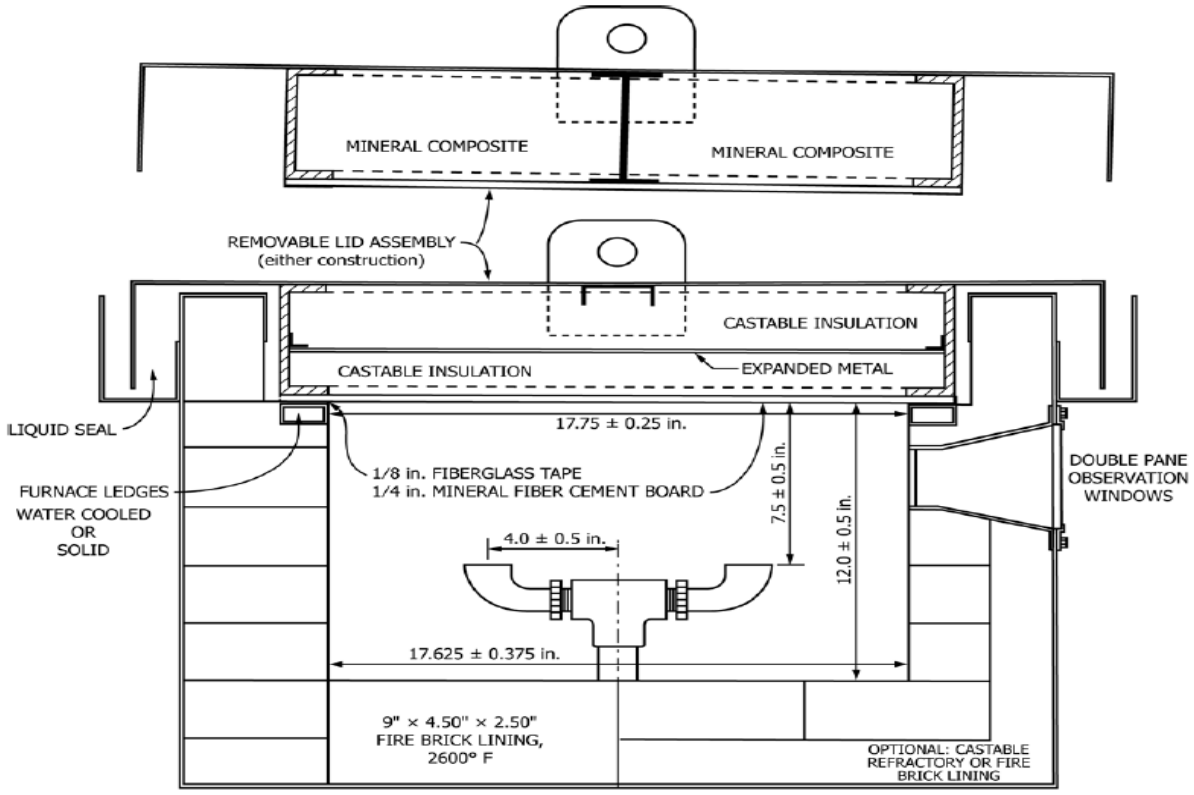


Diagram 2. Test Chamber looking down chamber showing critical dimensions.

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APPENDIX





Photos: Surface of Specimen Tested

REVISION HISTORY: Modifications have been made to the original issue from Monday, June 24, 2024

Per Matt Anderson with FlexTrim - Advised that the sample ID was incorrect. 062824
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*****<<<END OF TEST REPORT>>>*****

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